PRACTICAL LAB MANUAL

Web Development Student Guide

Three modular, beginner-friendly projects designed for VS Code & Live Server.

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1 Project 1: Light / Dark Mode Toggle

Objective: Learn how to use JavaScript to manipulate CSS classes dynamically to switch UI themes.

FILE SETUP & INSTRUCTIONS

Create a folder named `theme-toggle` and create three standard files inside it:

index.html — HTML Structure

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>Theme Toggle</title>
    <link rel="stylesheet" href="style.css">
</head>
<body>
    <div class="card">
        <h1>Theme Toggle</h1>
        <p>Click the button below to switch modes.</p>
        <button id="toggleBtn">Switch to Dark Mode</button>
    </div>
    <script src="script.js"></script>
</body>
</html>
```

style.css — CSS Styling

```
body {
    font-family: Arial, sans-serif;
    height: 100vh;
    display: flex;
    justify-content: center;
    align-items: center;
```

```
    background-color: #ffffff;
    color: #222222;
    transition: background-color 0.3s, color 0.3s;
}
body.dark-mode {
    background-color: #1e1e1e;
    color: #ffffff;
}
.card {
    text-align: center;
    padding: 30px;
    border: 1px solid #ccc;
    border-radius: 10px;
}
button {
    margin-top: 15px;
    padding: 10px 20px;
    font-size: 1rem;
    cursor: pointer;
    border: none;
    border-radius: 5px;
    background-color: #007bff;
    color: white;
}
```

script.js — JavaScript Logic

```
const btn = document.getElementById("toggleBtn");

btn.onclick = function() {
    document.body.classList.toggle("dark-mode");
    btn.textContent = document.body.classList.contains("dark-mode")
        ? "Switch to Light Mode"
        : "Switch to Dark Mode";
};
```

2 Project 2: Simple Click Counter

Objective: Understand numerical state tracking, incrementing, decrementing, and updating DOM values in real-time.

FILE SETUP & INSTRUCTIONS

Create a folder named `counter-app` with the three standard files:

index.html — HTML Structure

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>Click Counter</title>
    <link rel="stylesheet" href="style.css">
</head>
<body>
    <div class="box">
        <h2>Counter</h2>
        <h1 id="count">0</h1>
        <button onclick="updateCount(1)">+</button>
        <button onclick="updateCount(-1)">-</button>
        <button onclick="resetCount()">Reset</button>
    </div>
    <script src="script.js"></script>
</body>
</html>
```

style.css — CSS Styling

```
body {
    font-family: Arial, sans-serif;
    height: 100vh;
    display: flex;
    justify-content: center;
    align-items: center;
    background-color: #f3f4f6;
}
.box {
    background: white;
```

```
    padding: 30px;
    border-radius: 10px;
    text-align: center;
    box-shadow: 0 4px 10px rgba(0, 0, 0, 0.1);
}
h1 {
    font-size: 3rem;
    margin: 15px 0;
    color: #333;
}
button {
    padding: 10px 15px;
    font-size: 1rem;
    cursor: pointer;
    margin: 3px;
    border: none;
    border-radius: 5px;
    background-color: #28a745;
    color: white;
}
button:last-child {
    background-color: #dc3545;
}
```

script.js — JavaScript Logic

```
let num = 0;
const countEl = document.getElementById("count");

function updateCount(value) {
    num += value;
    countEl.textContent = num;
}

function resetCount() {
    num = 0;
    countEl.textContent = num;
}
```

3 Project 3: Show / Hide Password Input

Objective: Learn how to read and modify input element attributes dynamically based on user interaction.

FILE SETUP & INSTRUCTIONS

Create a folder named `password-toggle` with the three standard files:

index.html — HTML Structure

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>Show / Hide Password</title>
    <link rel="stylesheet" href="style.css">
</head>
<body>
    <div class="login-card">
        <h3>Enter Password</h3>
        <input type="password" id="password" placeholder="Enter password">
        <br>
        <button id="togglePass">Show Password</button>
    </div>
    <script src="script.js"></script>
</body>
</html>
```

style.css — CSS Styling

```
body {
    font-family: Arial, sans-serif;
    height: 100vh;
    display: flex;
    justify-content: center;
    align-items: center;
    background-color: #eef2f3;
}
.login-card {
    background: white;
    padding: 25px;
    border-radius: 8px;
```

```
    text-align: center;
    box-shadow: 0 4px 12px rgba(0,0,0,0.1);
}
input {
    padding: 10px;
    font-size: 1rem;
    margin: 15px 0;
    width: 200px;
    border: 1px solid #ccc;
    border-radius: 4px;
}
button {
    padding: 8px 15px;
    font-size: 0.9rem;
    cursor: pointer;
    border: none;
    background-color: #6c5ce7;
    color: white;
    border-radius: 4px;
}
```

script.js — JavaScript Logic

```
const passwordInput = document.getElementById("password");
const btn = document.getElementById("togglePass");

btn.onclick = function() {
    const isSecret = passwordInput.type === "password";
    passwordInput.type = isSecret ? "text" : "password";
    btn.textContent = isSecret ? "Hide Password" : "Show Password";
};
```

Generated for Student Web Development Labs. Use VS Code with Live Server extension to run these files.